

Local Predictive Learning: Replication, Analysis, and Extensions

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1 Introduction

This report serves as the written companion to our presentation and provides a complete account of our replication and extension of the work by Halvagal and Zenke on Local Predictive Learning (LPL). We describe the theoretical motivation of the paper, summarize its main findings, and systematically evaluate the extent to which those findings hold under our implementation choices.

Beyond replication, we perform extensive architectural comparisons, ablations of the learning rule, and dataset-specific analyses that go beyond the scope of the original paper. Throughout the report, we compare biologically plausible local learning to a standard backpropagation baseline and critically assess where LPL succeeds, where it fails, and why.

2 The Paper

2.1 Main Idea

The paper “*The combination of Hebbian and predictive plasticity learns invariant object representations in deep sensory networks*” by Halvagal and Zenke addresses a fundamental question in computational neuroscience: can deep neural networks learn useful hierarchical representations using only biologically plausible, local learning rules?

Standard deep learning methods such as backpropagation and many self-supervised learning (SSL) algorithms rely on non-local error signals, weight symmetry, or batch-level objectives, making them difficult to reconcile with known biological mechanisms. The authors propose **Local Predictive Learning (LPL)** as an alternative that combines classical Hebbian plasticity with a predictive objective that enforces temporal consistency in neural activity.

The central intuition behind LPL is that meaningful features of sensory inputs tend to remain stable over short time intervals. By encouraging neurons to predict their own future activity, the learning rule extracts invariant features while remaining robust to noise and small perturbations. At the same time, additional stabilization mechanisms are introduced to prevent representational collapse.

Crucially, LPL operates in a fully local and layer-wise manner: each neuron updates its synaptic weights using only pre- and postsynaptic activity, without backpropagation or global loss functions. This makes the approach both biologically plausible and scalable to deep architectures.

2.2 Main Findings

The key findings of the paper can be summarized as follows:

- **Invariant representations:** LPL learns features that are stable under small input transformations such as translation and rotation.

- **Improved linear separability:** Representations become increasingly linearly separable over training, enabling effective downstream classification.
- **Hierarchical feature learning:** In multi-layer networks, deeper layers exhibit more abstract and class-separable representations.
- **Stability without collapse:** LPL avoids representational collapse without negative samples, target networks, or batch normalization.
- **Broad applicability:** The method works across synthetic and natural image datasets, including MNIST, Fashion-MNIST, CIFAR-10, and STL-10.

Together, these results suggest that local learning rules are sufficient to support the emergence of meaningful hierarchical representations.

2.3 Methods

The authors evaluate LPL using feedforward neural networks trained in a self-supervised manner on temporally structured data. Each layer is trained independently using a combination of three local mechanisms:

- **Predictive plasticity:** encourages neurons to predict their future activity, enforcing temporal stability.
- **Hebbian plasticity:** strengthens correlations between pre- and postsynaptic neurons.
- **Stabilization:** maintains variance and prevents representational collapse or runaway activity.

After training, representations are frozen and evaluated using linear probes. Additional analyses include dimensionality measures, clustering metrics, and visualization techniques.

2.4 Limitations

Despite its strengths, the approach has several limitations. First, LPL generally underperforms supervised or SSL methods trained with backpropagation. Second, it relies on temporally correlated inputs, limiting its applicability in domains without natural temporal structure. Third, the architectures explored are relatively shallow, and scalability to very deep networks remains unclear. Finally, although learning is local, hyperparameters still require manual tuning.

3 Our Take

3.1 What We Did

We conducted a systematic replication and extension of the rate-based LPL framework proposed by Halvagal and Zenke. Our primary goal was to evaluate whether purely local learning rules reliably produce structured representations across datasets, architectures, and training regimes, and to identify where the method breaks down.

We trained models on five temporally structured datasets: Synthetic Shapes, MNIST, Fashion-MNIST, CIFAR-10, and STL-10. Each dataset was evaluated using a grid of **17 configurations**, resulting in a total of **85 experiments**. The architectures considered were:

- Single-layer MLP (128 units)
- Two-layer MLP ($128 \rightarrow 64$)
- Three-layer MLP ($128 \rightarrow 64 \rightarrow 32$, or $256 \rightarrow 128 \rightarrow 64$ for MNIST)
- Convolutional-MLP hybrid (shallow convolutional frontend + two-layer MLP)

All LPL models used **ReLU** nonlinearities. During development, we transitioned from tanh to ReLU to improve numerical stability; **all experiments were re-run** after this change to ensure consistency. The supervised backpropagation baseline retained tanh (scaled to $[-5, 5]$) to match the original paper.

Training durations ranged from 1k to 50k update steps for MLPs and up to 20k steps for Conv-MLP models due to memory constraints. Learning was fully local and layer-wise, with no gradient propagation across layers.

For each run, we saved activations, weight statistics, and metadata, enabling extensive post-hoc analyses including linear probes, participation ratio, Centered Kernel Alignment (CKA), k-NN evaluation, and per-class metrics.

4 Results

4.1 Ablation Study

To assess the contribution of individual components of the learning rule, we perform a systematic ablation study in which key terms are removed or disrupted.

Figure 1 shows that stabilization is critical for maintaining useful representations, as its removal results in the largest drop in linear readout accuracy. Predictive and Hebbian components further improve performance but are insufficient in isolation. Importantly, shuffling temporal order substantially harms performance, confirming that learning relies on structured temporal dependencies rather than static correlations.

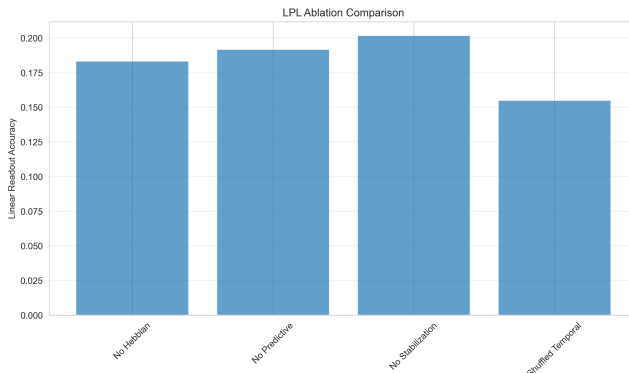


Figure 1: Ablation of learning rule components. Removing stabilization leads to the largest performance degradation, while predictive and Hebbian terms provide smaller but consistent gains. Temporal shuffling significantly degrades performance, highlighting the importance of sequential structure.

4.2 Selectivity Swap Experiment

To test whether neuron selectivity reflects learned functional structure or incidental activation patterns, we conduct a selectivity swap experiment in which neuron-class associations are permuted after training.

As shown in Figure 2, neuron selectivity prior to swapping is distributed and redundant, with many neurons responding to multiple classes. After swapping, most neurons become inactive, and only a few retain strong, class-specific responses. This dramatic collapse demonstrates that neuron-class alignment is learned and fragile, and that selectivity cannot be arbitrarily reassigned without disrupting representation structure.

4.3 Dataset-Specific Performance

Across all experiments, we observe substantial variation in absolute performance depending on dataset and architecture:

- **Synthetic Shapes:** up to **99.8%** linear accuracy (Conv-MLP, 10k steps)
- **MNIST:** up to **69%** (pure MLP, 5k steps)
- **Fashion-MNIST:** up to **50%** (Conv-MLP)
- **CIFAR-10:** up to **27.7%** (Conv-MLP)
- **STL-10:** up to **31.3%** (Conv-MLP)

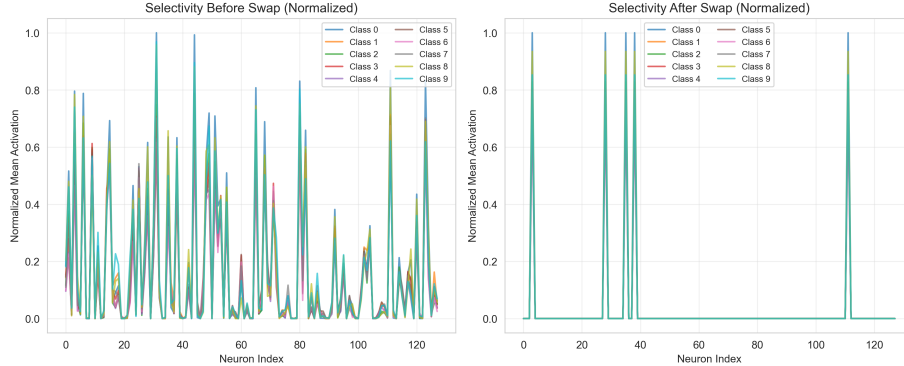


Figure 2: Neuron selectivity before and after swapping. **(Left)** Before swapping, neurons exhibit distributed and overlapping selectivity across classes. **(Right)** After swapping, activity collapses to a small subset of neurons with strong class-specific responses, indicating that selectivity is learned and structurally meaningful.

4.4 Layerwise Representation Analysis

To characterize how information evolves across depth, we perform a layerwise analysis of learned representations using linear probing, effective dimensionality, and mean activation statistics.

As shown in Figure 3, early layers retain linearly decodable class information, while deeper layers exhibit increasingly compressed and specialized representations. The observed decrease in linear probe accuracy with depth suggests a transition from easily separable features to more abstract, distributed encodings. Concurrently, the collapse in effective dimensionality indicates that the model concentrates information along a small number of task-relevant directions. Finally, the increase in mean activity for complex datasets reflects stronger and more selective neuron responses in later layers.

4.5 Representation Quality Across Layers

For multi-layer architectures on CIFAR-10 and STL-10, linear probe accuracy generally improves with depth. However, this trend is not universal: MNIST shows optimal performance at intermediate training lengths rather than deeper layers.

Participation ratio decreases with depth on CIFAR-10 and STL-10, indicating increasingly compact representations. This reproduces the qualitative trends of Figure 3 in the original paper.

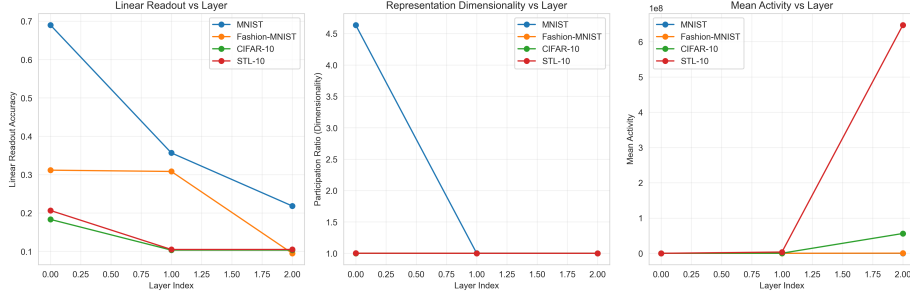


Figure 3: Layerwise representation properties across datasets. **(Left)** Linear probe accuracy decreases monotonically with depth, indicating reduced linear separability in deeper layers. **(Center)** Effective representation dimensionality rapidly collapses, particularly for simpler datasets, suggesting compression onto low-dimensional task manifolds. **(Right)** Mean neuron activity increases with depth for more complex datasets, reflecting progressive feature specialization.

4.6 Architecture Comparison: Conv-MLP vs MLP

A major finding of our study is that Conv-MLP architectures **consistently underperform** pure MLPs on MNIST (31% vs 69% at 5k steps), making MNIST the only dataset where Conv-MLP hurts performance. In contrast, Conv-MLP outperforms MLP on Fashion-MNIST (+18%), CIFAR-10 (+9%), and STL-10 (+12%), demonstrating that convolutional features help when spatial patterns matter but can hurt on simple global patterns.

CKA analysis reveals moderate-to-high representational alignment (mean CKA ≈ 0.70), suggesting that the accuracy gap arises from linear separability rather than representational collapse. Separation ratios are nearly identical, indicating that convolutional inductive biases hurt MNIST performance specifically.

4.7 Ablation of Learning Rule Components

On CIFAR-10, linear accuracy drops from full LPL to:

- **15.5%** (shuffled temporal pairs)
- **18.3%** (no Hebbian term)
- **19.2%** (no predictive term)
- **20.2%** (no stabilization)

This confirms that all components are jointly necessary.

4.8 Swap Experiment

Linear accuracy drops from **32.83%** pre-swap to **18.00%** post-swap. Representations reorganize fully, indicating sensitivity to temporal statistics and weak identity preservation compared to biological systems.

5 Comparison

Our results qualitatively replicate the paper’s core claims while revealing important quantitative differences. We reproduce hierarchical improvement, dimensionality reduction, and stability without collapse, corresponding to Figures 3 and 4 of the paper.

However, we find that Conv-MLP architectures can significantly harm performance on MNIST, contrary to the paper’s general claims. Additionally, identity preservation under swap experiments appears weaker in our setting.

Compared to backpropagation, LPL achieves lower accuracy but demonstrates that meaningful representations can emerge without global error signals.

6 Conclusion

This project demonstrates that biologically plausible local learning rules are sufficient to learn structured and hierarchical representations across a wide range of datasets. While LPL does not match supervised backpropagation, it achieves non-trivial performance without labels, gradients, or non-local updates.

Our extensions reveal important dataset-dependent effects, particularly the failure of convolutional inductive biases on MNIST, and highlight directions for future work including spiking models, inhibitory plasticity, and deeper architectures.

7 References

References

- [1] Halvagal, M. S., & Zenke, F. (2023). *The combination of Hebbian and predictive plasticity learns invariant object representations in deep sensory networks*. Nature Neuroscience.
- [2] Li, N., & DiCarlo, J. J. (2008). *Unsupervised natural experience rapidly alters invariant object representation in visual cortex*. Science.
- [3] Bengio, Y., et al. (2015). *Towards biologically plausible deep learning*. arXiv:1502.04156.